



Algebra and Functions · Worksheet 1

Chapter 2: Algebra and Functions. Answer all questions. Show your working clearly. Total: 184 marks. Time: 3 hours 50 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are nine sections, one for each topic in Chapter 2. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Give answers in exact form (fractions, surds) unless the question says otherwise.
- When a question says “use algebra”, solutions relying entirely on calculator technology are not acceptable.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Indices and surds	16	
B Quadratics	17	
C Simultaneous equations	24	
D Inequalities	20	
E Factor and remainder theorems	21	
F Partial fractions	21	
G Graph transformations	19	
H Functions: domain, range, composite and inverse	22	
I The modulus function	20	
Roar-level challenge	4	
Total	184	

Section A Indices and surds

16 marks

1. Find the exact value of $27^{-\frac{2}{3}}$.

[2]

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2. Simplify $\sqrt{75} - \sqrt{12} + \sqrt{48}$, giving your answer in the form $k\sqrt{3}$, where k is an integer. [3]

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3. Express $\frac{6}{\sqrt{5} - \sqrt{2}}$ in the form $a\sqrt{5} + b\sqrt{2}$, where a and b are integers. [3]

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4. Write $\frac{x^2 + 4}{2\sqrt{x}}$ in the form $px^m + qx^n$, where p , q , m and n are constants. [3]

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5. Given that $y = 2^x$,

(a) express each of 4^x and 2^{x+3} in terms of y .

[2]

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(b) Hence, using algebra, solve the equation

[3]

$$4^x + 2^{x+3} = 20.$$

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(Total for Question 5 is 5 marks)



Section B Quadratics

17 marks

6. Solve the equation $2x^2 - 7x + 3 = 0$.

[2]

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7. (a) Write $x^2 + 8x + 5$ in the form $(x + p)^2 + q$, where p and q are integers.

[3]

(b) Hence state the coordinates of the minimum point of the curve $y = x^2 + 8x + 5$.

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8. Calculate the value of the discriminant of $3x^2 - 4x + 5$, and hence state the number of real roots of the equation $3x^2 - 4x + 5 = 0$. Give a reason for your answer.

[3]

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9. (a) Express $5 + 12x - 3x^2$ in the form $a - b(x + c)^2$, where a , b and c are integers.

[3]

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(b) Hence state the maximum value of $5 + 12x - 3x^2$ and the value of x at which it occurs.

[1]

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(Total for Question 9 is 4 marks)

10. The equation $kx^2 + 4kx + 3 = 0$, where k is a constant, has no real roots.

[5]

Find the set of possible values of k .

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Section C Simultaneous equations

24 marks

11. Solve the simultaneous equations

[3]

$$3x + 2y = 7, \quad 5x - 4y = 19.$$

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12. Solve the simultaneous equations

[4]

$$y = x + 1, \quad y = x^2 - 5.$$

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13. Use algebra to solve the simultaneous equations

[5]

$$x - y = 4, \quad x^2 + xy = 6.$$

14. The line with equation $y = 2x + k$, where k is a constant, is a tangent to the curve with equation $y = x^2 + 3$.

(a) Find the value of k .

[3]

(b) Find the coordinates of the point where the line touches the curve.

[2]

(Total for Question 14 is 5 marks)



15. Use algebra to solve the simultaneous equations

[7]

$$x + y = 2, \quad 4y^2 - x^2 = 11.$$

You must show all stages of your working.

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Section D Inequalities

20 marks

16. Solve the inequality $4x - 7 < 2x + 5$.

[2]

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17. Solve the inequality $3(2 - x) \geq 5x + 22$.

[3]

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18. Solve the inequality $2x^2 + 5x - 3 > 0$.

[4]

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19. Find the set of values of x for which

(a) $3x - 5 > x + 1$,

[1]

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(b) $x^2 - 7x + 10 < 0$,

[3]

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(c) both $3x - 5 > x + 1$ and $x^2 - 7x + 10 < 0$.

[1]

Give your answer in set notation.

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(Total for Question 19 is 5 marks)



20. The line l has equation $y = kx - 2$, where k is a constant.

The curve C has equation $y = x^2 + 2x + 2$.

Given that l and C do **not** intersect,

(a) show that $(2 - k)^2 < 16$,

[3]

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(b) hence find the range of possible values of k .

[3]

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(Total for Question 20 is 6 marks)



Section E Factor and remainder theorems

21 marks

21. Find the remainder when $f(x) = 2x^3 - 3x^2 + x - 5$ is divided by $(x - 2)$. [2]

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22. Use the factor theorem to show that $(x + 3)$ is a factor of $x^3 + 2x^2 - 5x - 6$. [2]

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23. Factorise completely $x^3 - 7x - 6$. [4]

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24. $f(x) = x^3 + ax^2 + bx + 6$, where a and b are constants.

[5]

Given that $(x - 1)$ is a factor of $f(x)$, and that when $f(x)$ is divided by $(x + 1)$ the remainder is 8, find the value of a and the value of b .

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25.

$$f(x) = 2x^3 - 5x^2 - 4x + 3$$

- (a) Use the factor theorem to show that $(x - 3)$ is a factor of $f(x)$. [2]

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- (b) Hence factorise $f(x)$ completely. [4]

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- (c) Hence solve the equation $f(x - 1) = 0$. [2]

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(Total for Question 25 is 8 marks)



Section F Partial fractions

21 marks

26. Express $\frac{5x+1}{(x+1)(x-1)}$ in partial fractions.

[3]

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27. Express $\frac{x+10}{(x+3)(2x-1)}$ in partial fractions.

[4]

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28. Express $\frac{2x^2+6x-2}{x(x-1)(x+2)}$ in partial fractions.

[4]

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29. Given that

[5]

$$\frac{x^2 + 7x + 1}{(x-1)^2(x+2)} \equiv \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2},$$

find the values of the constants A , B and C .

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30.

[5]

$$\frac{3x^2 + 2x - 2}{(x+2)(x-1)} \equiv A + \frac{B}{x+2} + \frac{C}{x-1}$$

Find the values of the constants A , B and C .

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Section G Graph transformations

19 marks

31. The curve $y = f(x)$ passes through the point $(2, -6)$. [2]

Write down the coordinates of the corresponding point on each of the curves

(a) $y = f(x) + 4$,

(b) $y = f(x + 3)$.

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32. The curve $y = f(x)$ has a maximum point at $(-4, 6)$. [3]

Write down the coordinates of the maximum or minimum point on each of the following curves.

(a) $y = 3f(x)$

(b) $y = f\left(\frac{1}{2}x\right)$

(c) $y = -f(x)$

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33. Describe fully the single geometrical transformation that maps [3]

(a) the curve $y = x^2$ onto the curve $y = (x - 5)^2$,

(b) the curve $y = x^3$ onto the curve $y = 8x^3$. (Give **two** possible answers.)

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34. The curve $y = x^2$ is translated by the vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$.

(a) Find the equation of the translated curve, giving your answer in the form $y = x^2 + bx + c$. [2]

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(b) The translated curve is then stretched parallel to the y-axis with scale factor 2. [2]
Find the equation of the final curve.

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(Total for Question 34 is 4 marks)



35. Figure 1 shows a sketch of the curve with equation $y = f(x)$.

The curve passes through the origin O and the point $(4, 0)$, and has a minimum point at $(2, -4)$.

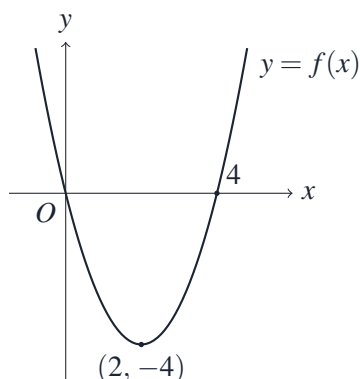


Figure 1

On separate diagrams, sketch the curves with the following equations. On each sketch, show clearly the coordinates of the turning point and of any points where the curve meets the coordinate axes.

(a) $y = f(x+2)$

[3]

(b) $y = 2f(x)$

[2]

(c) The curve $y = f(x) + k$ touches the x -axis. State the value of the constant k .

[1]

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(d) Given that $f(x) = x^2 - 4x$, find an equation for the curve $y = f(x+2)$ in its simplest form.

[1]

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(Total for Question 35 is 7 marks)



Section H Functions: domain, range, composite and inverse

22 marks

36. The functions f and g are defined by

[2]

$$f(x) = 4x - 3, \quad g(x) = x^2 + 1, \quad x \in \mathbb{R}.$$

Find (a) $fg(2)$, (b) $gf(1)$.

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37. The function f is defined by

[3]

$$f(x) = \frac{2x+1}{x-3}, \quad x \in \mathbb{R}, x \neq 3.$$

Find $f^{-1}(x)$, stating its domain.

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38. State the range of each of the following functions.

[3]

(a) $f(x) = x^2 + 2, \quad x \in \mathbb{R}$

(b) $g(x) = 3 - 2x, \quad -1 \leq x \leq 4$

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39. The function f is defined by

$$f(x) = x^2 - 6x + 11, \quad x \geq 3.$$

(a) Express $f(x)$ in the form $(x - a)^2 + b$, where a and b are constants.

[2]

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(b) Hence state the range of f .

[1]

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(c) Find $f^{-1}(x)$, stating its domain.

[3]

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(Total for Question 39 is 6 marks)



40. The functions f and g are defined by

$$f(x) = 2x + 5, \quad x \in \mathbb{R}, \quad g(x) = \frac{3}{x-1}, \quad x > 1.$$

(a) State the range of g . [1]

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(b) Find $g^{-1}(x)$, stating its domain. [3]

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(c) Solve the equation $fg(x) = x$, giving your answer in the form $p + \sqrt{q}$, where p and q are integers. [4]
Justify your answer.

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(Total for Question 40 is 8 marks)



Section I The modulus function

20 marks

41. Solve the equation $|x - 4| = 7$.

[2]

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42. Solve the inequality $|2x + 3| < 5$.

[3]

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43. Solve the inequality $|3x - 2| \geq 4$.

[3]

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44. Solve the equation $|x + 5| = |3x - 1|$.

[4]

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45. (a) On the same axes, sketch the graphs of $y = |2x - 6|$ and $y = x + 1$. [2]
Show the coordinates of any points where each graph meets the coordinate axes.

- (b) Use algebra to solve the equation $|2x - 6| = x + 1$. [4]

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- (c) Hence, or otherwise, find the set of values of x for which $|2x - 6| > x + 1$. [2]

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(Total for Question 45 is 8 marks)



ROAR-LEVEL CHALLENGE

4 marks

46. Solve the equation $|x^2 - 5| = 4$.

[4]

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END OF WORKSHEET

Total: 184 marks